

Table 1. 2022 independent-test performance

No	Input	Random Forest MAE	Random Forest RMSE	Random Forest PCC	XGBoost MAE	XGBoost RMSE	XGBoost PCC	LightGBM MAE	LightGBM RMSE	LightGBM PCC	CatBoost MAE	CatBoost RMSE	CatBoost PCC
1	Comb1	63.24	92.75	0.732	67.52	95.49	0.706	66.48	95.20	0.693	64.79	98.08	0.689
2	Comb1_land	55.95	86.97	0.744	58.63	89.89	0.716	63.70	99.35	0.644	53.27	85.98	0.756
3	Comb2	67.79	96.02	0.750	64.32	100.80	0.706	56.86	81.26	0.825	60.62	92.43	0.770
4	Comb2_land	57.48	89.18	0.769	63.69	96.57	0.731	61.47	91.18	0.759	54.75	87.91	0.771
5	Comb3	63.91	98.41	0.707	66.03	98.74	0.709	68.71	100.78	0.698	67.20	97.84	0.709
6	Comb3_land	57.43	88.94	0.740	57.76	92.49	0.718	63.44	95.28	0.703	59.30	90.22	0.733
7	CCS_land	57.88	93.07	0.709	61.71	95.49	0.700	63.21	97.56	0.685	62.29	94.68	0.702
8	PDIR_land	56.32	88.94	0.740	60.17	95.86	0.694	63.60	95.60	0.700	58.98	91.17	0.730
9	GSMaP_MVK_land	54.50	98.24	0.638	61.53	105.94	0.580	63.44	104.52	0.593	56.69	98.66	0.646
10	CDR_land	58.51	95.95	0.693	69.25	110.58	0.591	66.50	106.98	0.622	60.35	98.37	0.676
11	CHIRPS_land	56.38	88.95	0.751	61.12	94.15	0.731	61.48	94.87	0.726	54.31	85.66	0.782
12	IMERG_land	55.41	85.69	0.759	64.16	97.30	0.686	60.73	94.03	0.709	57.24	87.63	0.752
13	GSMaP_Gauge_land	52.14	85.76	0.792	54.00	87.79	0.780	57.21	88.34	0.783	56.38	90.06	0.786
14	ERA5_land	51.83	84.92	0.770	59.46	98.11	0.687	57.11	97.01	0.701	55.73	87.76	0.756